Lesson 2: Fields

ChatGPT edition — staged field mapping

A map of force, built from repeated tests

Stop line and roles

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

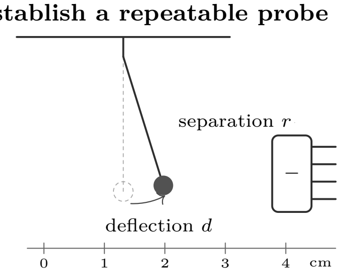
What you need

- Tape charge source
- Thread-suspended pith or foil ball
- Ruler and graph paper
- Fine pencil, eraser, and timer
- Two books or blocks to fix the support

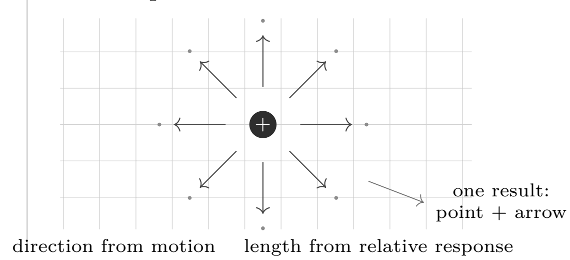
Predict first

As separation doubles, will the interaction strengthen, weaken, or remain unchanged? Draw the arrow you expect at three different locations. What observation would prove your prediction wrong?

1. Establish a repeatable probe



2. Sample one location at a time



What the drawing claims

A field map assigns an interaction to each location. At every sampled point, the arrow records two claims: its direction shows how a positive test charge would initially accelerate, and its length represents relative response under the same test conditions. A map is therefore inferred from repeated trials, not copied from something directly visible.

Checkpoint before measuring

If the ball, thread length, charge source, or starting position changes between trials, arrow lengths can no longer be compared. Name the four controls aloud. The adult checks that the source never touches the suspended probe.

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ChatGPT edition — collect the evidence

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Stage 1: make the probe reproducible

1. Suspend the light ball. Shield it from drafts and wait until it rests. Mark the rest position on graph paper behind it.
2. Choose one source position and perform three approaches without contact. If the resting point drifts, stop: discharge/reset the materials and repeat.
3. Decide how deflection will be read: horizontal displacement d, not the curved travel of the ball. Always read the ruler from the same angle.

Intermediate questions

1. Does the ball move toward or away from the source on the first approach?
2. Does it return to the same rest mark? If not, which variable changed?
3. Would a heavier ball give the same deflection? Would it indicate the same field direction? Explain the difference.

Stage 2: measure distance and response

Keep the source orientation, ball, thread length, and approach time fixed. Measure r from the same reference points. Use at least five separations and repeat every reading.

Trial	r (cm)	d_1 (mm)	d_2 (mm)	mean d (mm)	Direction / condition
1					
2					
3					
4					
5					

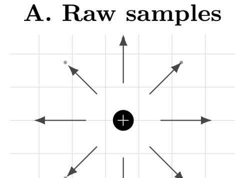
Question at the halfway point

Compare the nearest and farthest readings before continuing. Is the sign of the direction consistent? Is the size trend larger than the repeat-to-repeat variation? If not, collect another repeat instead of forcing a conclusion.

Stage 3: turn trials into a map

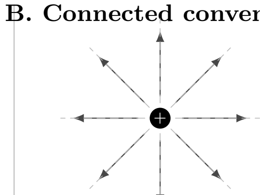
1. Mark each tested location as a dot on graph paper.
2. At the dot, draw an arrow in the observed direction. Choose one scale, such as 1 cm of arrow per 1 mm of mean deflection.
3. Repeat around the source. Only after the arrows are plotted may you add light guide curves through arrows that already agree.

A. Raw samples



Keep measured variation visible.
Do not "repair" data by eye.

B. Connected convention



Lines summarize the arrows;
they are not tracks or strands.

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Worked inference

Suppose the repeated measurements at $r = 2\,\text{cm}$ are $d_1 = 8\,\text{mm}$ and $d_2 = 10\,\text{mm}$, while at $r = 4\,\text{cm}$ they are $d_1 = 2\,\text{mm}$ and $d_2 = 3\,\text{mm}$.

From numbers to a defensible claim

$$\bar{d}_{2\text{cm}} = (8+10)/2 = 9\,\text{mm}, \qquad \bar{d}_{4\text{cm}} = (2+3)/2 = 2.5\,\text{mm}.$$

Doubling separation reduced the response to $2.5/9 \approx 0.28$, roughly one quarter. The evidence supports "the interaction weakened strongly" for this apparatus. It does *not* by itself prove an exact inverse-square law: deflection is an indirect response affected by thread geometry, probe mass, charge leakage, and reading uncertainty.

Explain from the evidence

An electric-field arrow points in the direction a positive test charge would accelerate. Field lines are a drawing convention: they do not appear because ink connects arrows. Lines cannot cross, because a crossing would assign two directions to the same location. Closer or longer arrows represent a stronger relative response only when the probe and scale remain unchanged.

Diagnostics

Symptom	Likely cause	Correction
Rest point creeps	charge leakage, draft, twisted thread	reset, shield, wait
Direction reverses after contact	source touched probe; charge transferred	discharge/reset; approach without contact
Repeats disagree widely	ruler parallax or inconsistent timing	fix viewing point; use the same pause
All arrows equal	scale rounded too coarsely	record millimetres; enlarge drawing scale
Guide lines cross	curves were invented between sparse samples	erase curves; add measurements

Final verification

1. Point to one arrow and state the measurement that fixes its direction and the calculation that fixes its length.
2. Find the largest disagreement between repeats. Mark that arrow with an uncertainty bracket or collect a third trial.
3. Cover the source label. Can another reader infer where the source lies from the arrows alone? Ask them; record their answer.
4. Redraw the prediction with a dashed line over the measured map. Circle one agreement and one disagreement, then write what new test would distinguish two explanations.

Evidence record

Prediction	__________	Controls	__________
Key observation	__________	Claim + limitation	__________

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.